Quiz 5

ECON 320 – Introduction to Mathematical Economics

Instructor: Muhebullah Karimzada

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Multiple Choice Questions (1 mark each)

1. The **Hessian matrix** of a function $f(x, y)$ is:
   (a) A matrix of first partial derivatives
   (b) A matrix of second partial derivatives
   (c) A matrix of total differentials
   (d) The Jacobian of f

2. If $D = f_{xx}f_{yy} - (f_{xy})^2 > 0$ and $f_{xx} < 0$, the point (x^*, y^*) is:
   (a) A saddle point
   (b) A local maximum
   (c) A local minimum
   (d) An inflection point

3. For $f(x, y) = x^2 - 2xy + 3y^2$, the determinant of the Hessian D is:
   (a) $D = -8$
   (b) $D = 0$
   (c) $D = 2$
   (d) $D = 8$

4. The second-order condition for a local maximum of $f(x, y)$ is satisfied when:
   (a) $D > 0$ and $f_{xx} > 0$
   (b) $D < 0$
   (c) $D > 0$ and $f_{xx} < 0$
   (d) $f_{xy} = 0$

5. Suppose $f(x, y) = 4x^2 + 9y^2 - 12xy$. Which of the following statements is **true**?
   (a) The function has a maximum at $(0, 0)$
   (b) The function has a minimum at $(0, 0)$
   (c) The function has a saddle point at $(0, 0)$
   (d) The determinant of the Hessian is zero at $(0, 0)$

Problem 1 (6 marks)
Let $f(x, y) = 3x^2 + xy + y^2 - 6x - 9y$.

(a) Find all stationary points.
(b) Determine their nature (maximum, minimum, or saddle) using the Hessian determinant test.

Problem 2 (9 marks)
A firm's production function is $Q = f(K, L) = 100K^{0.5}L^{0.5}$. The cost of capital is $r = 4$ and the wage rate is $w = 1$. The firm wants to produce 100 units of output.

(a) Formulate the Lagrangian and find the first-order conditions.
(b) Solve for the cost-minimizing values of K and L.
(c) Compute the minimum cost.
(Hint: use the property that at optimum, $\frac{MPL}{MPK} = \frac{w}{r}$.)

End of Quiz